KAAN GULER

AI / Deep Learning Engineer

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PROFESSIONAL SUMMARY

Computer Engineering graduate (High Honors) and AI engineer with hands-on PyTorch experience across real-time computer vision and production deep-learning systems. Led the perception sub-team of a funded autonomous-vehicle research project, delivering low-latency object detection and semantic segmentation models optimized for resource-constrained edge devices through GPU/CUDA-aware optimization and model quantization. Defense-industry internship experience (HAVELSAN) developing core backend systems in C++. Strong foundation in CNNs, RNNs, Transformers, and attention mechanisms, complemented by four end-to-end LLM/agentic products demonstrating reasoning-oriented orchestration and real-time streaming inference. Motivated to apply modern deep learning to neural video/audio compression, adaptive codecs, and real-time edge AI for next-generation communication systems.

TECHNICAL SKILLS

Programming: Python (primary), C++, C#, Java, SQL, JavaScript

Deep Learning & Model Development: PyTorch, CNNs, RNNs, Transformers, Attention Mechanisms, Sequence Modeling, Hugging Face Transformers, Model Training, NLP

Edge AI & Optimization: Model Quantization, Inference Optimization, GPU/CUDA Programming, Low-Latency & Real-Time Inference, Edge Deployment, vLLM, Ollama

Computer Vision: Object Detection, Semantic Segmentation, Real-Time Perception, OpenCV, Edge Inference

LLM & Agentic Systems: Frontier LLMs (OpenAI, Gemini, Anthropic), Reasoning & Agentic Workflows, RAG, Tool Calling, Structured Outputs, Model Context Protocol (MCP), LangGraph, LangChain, LlamaIndex, Prompt Engineering, LLM Evaluation & Benchmarking, Streaming Completions

MLOps & Infrastructure: Docker, GitHub Actions (CI/CD), MLflow, Optuna, Git, Unix/Linux, FastAPI (async)

Data & Vector Stores: NumPy, Pandas, Scikit-Learn, PostgreSQL, Redis, SQLite, Qdrant, ChromaDB, FAISS, EDA, Feature Engineering

RESEARCH EXPERIENCE

Computer Vision Sub-Team Lead | Autonomous Vehicle (Funded Research, TEDU) *Sep 2024 – Jun 2025*

- Led a computer-vision engineering team building the real-time perception sub-system of an autonomous scale vehicle, owning model development, data pipelines, and on-vehicle integration.
- Developed object detection and semantic segmentation models in PyTorch and OpenCV, optimized for low-latency edge inference on resource-constrained hardware.
- Applied GPU/CUDA-aware optimization and model quantization to cut inference latency and memory footprint while preserving accuracy for real-time operation.
- Built an end-to-end ML platform with MLflow (experiment tracking, model versioning) and Optuna (automated hyperparameter optimization), establishing reproducibility and production-monitoring practices.

PROJECTS

CurriculumForge | Autonomous Multi-Agent Reasoning System (Grounded LLM Pipeline) *Apr 2026 – May 2026*

Python, LangGraph, ChromaDB, Pydantic, SQLite, Ollama-hosted LLaMA / Gemini / Groq

- Architected a 5-agent LangGraph reasoning pipeline (selector, generator, reviewer, refiner) producing grounded, schema-validated outputs from minimal input, applying multi-step reasoning orchestration to a complex generation problem.
- Ran open-source models locally via Ollama-hosted LLaMA alongside hosted providers behind a provider-agnostic abstraction, with versioned prompts and a self-correcting reviewer-feedback retry loop.
- Designed a 4-layer deterministic evaluation framework cross-checking every generation against a normalized reference index; covered by 115 passing pytest tests and reproducible benchmark runs.

TestForge | Multi-Agent AI Test Generation System *Jan 2026 – Mar 2026*

Python, LangGraph, OpenAI API, Docker, GitHub Actions, Pydantic

- Built an agentic reasoning system (writer, runner, analyzer, suggester) that autonomously generates, executes, and analyzes software tests, applying LLM reasoning to an end-to-end engineering workflow.
- Implemented self-correcting agent loops with failure classification to recover from tool-call errors and improve execution reliability.
- Integrated Docker-based sandboxing and a GitHub Actions CI/CD pipeline for safe execution, plus a custom benchmarking framework tracking per-agent performance across runs.

Pressing Analyst Bot | Real-Time Streaming RAG Assistant *Feb 2026 – May 2026*

Forked from TwelveGPT Educational · Python, OpenAI API (Tool Calling), Cosine-Similarity Embeddings, Streaming Responses

- Engineered real-time, low-latency streaming completions for token-by-token UX over a grounded analytics dataset.
- Designed the context-engineering layer (structured prompt templates, tool-call schemas, and function-calling integration) so the model queries metrics on demand instead of free-form generation.
- Reinforced the pipeline with cosine-similarity retrieval and custom guardrails to keep responses grounded and on-topic.

InsightMail | Local-First, On-Premise AI Assistant (FastAPI + RAG) *Nov 2025 – Jan 2026*

Python, FastAPI (async), LLaMA via Ollama, Qdrant, Sentence Transformers, IMAP

- Built a privacy-preserving, fully on-premise assistant running locally hosted LLM inference; sensitive data never leaves the local environment (KVKK/GDPR-aligned), a pattern directly relevant to secure, classified deployments.
- Engineered the full pipeline as a concurrent async FastAPI service: IMAP ingestion, Sentence-Transformer embeddings, Qdrant vector indexing, semantic retrieval, and local generation.
- Designed sentence-aware chunking and dynamic context assembly with cosine-similarity retrieval for accurate long-context responses.

PROFESSIONAL EXPERIENCE

AI / Backend Engineer (Freelance) Remote / Bolu, Turkey *Jun 2025 – Present*

- Contributed backend and AI features to a production mobile application as an external developer, working with the core team across release cycles.
- Designed and integrated LLM-powered features (semantic search, a conversational assistant, and smart suggestions) through async FastAPI services calling OpenAI and Gemini APIs.
- Built and maintained REST endpoints consumed by the mobile client, tuning response latency, error handling, and clean API contracts.
- Pursued self-directed study in agentic architectures (LangGraph) and MLOps, applied across personal AI projects.

Usta Bilgi Sistemleri | Computer Engineer Intern Ankara, Turkey *Aug 2024 – Sep 2024*

- Built backend components for a corporate document-signing application using .NET, React, and PostgreSQL.
- Implemented coordinate-based canvas positioning and image-processing routines for digital signature placement; containerized runtimes with Docker for predictable deployment.

HAVELSAN | Computer Engineer Intern Ankara, Turkey *Jul 2024 – Aug 2024*

- Contributed to core backend system development in a defense-industry environment, working in C++ alongside a JavaScript / Node.js stack (Express.js, Vue.js).
- Built robust, enterprise-grade code following unit-testing methodologies, integrating native C++ modules into Node.js services via Koffi (foreign-function interface).

Ziraat Technology | Computer Engineer Intern Ankara, Turkey *Jul 2023 – Sep 2023*

- Developed RESTful APIs and MVC architectures for regulated financial applications using ASP.NET Core.
- Refactored relational SQL queries to reduce latency and increase pipeline capacity, under financial-sector compliance constraints.

EDUCATION

TED University | B.Sc. in Computer Engineering *Sep 2020 – Jun 2025*

GPA: 3.60 / 4.00 (High Honors)

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Systems, Software Engineering, Computer Networks, Operating Systems, Linear Algebra, Probability & Statistics.

Leadership: TEDU IEEE, Unity Mentor for 8 teams; Game Dev Society, Supervisory Board; AlecTed Team, Shell Eco-Marathon autonomous-vehicle software.

LANGUAGES & ADDITIONAL

Languages: Turkish (Native); English (Professional Proficiency, IELTS 6.5).

Interests: Basketball (10 years, school & club); Korfball (provincial champion). **References:** Available upon request.